

5

Magnetic Fields

In this chapter, we will learn about

- the forces and magnetic fields produced by magnets
- the magnetic fields produced by currents
- electromagnets

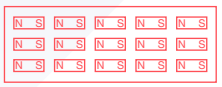
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5.1 Magnetic forces and magnetic fields

For simplicity, a magnetic material can be regarded as composing of many randomly orientated tiny magnets.



When the material is magnetized, the tiny magnets are lined up and the material as a whole behaves like a magnet.



DSE goal

Realize the attraction and repulsion between magnetic poles.

Flipped classroom

Watch a video and answer the questions.

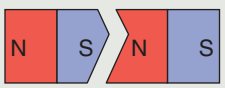
May stress that not all metals can be attracted by magnets.

Some rare earth metals such as gadolinium, samarium, neodymium are strongly magnetic.

Know more

Broken magnet

If a magnet is broken in the middle, we get two smaller magnets, each still having two poles. It seems that poles always exist in pairs. For centuries, scientists have searched for a magnet with a single pole, but without success.



Have you ever wondered why a refrigerator door can close tightly? The answer is simple. There are magnets inside the seal of the door so that the door is attracted to the steel body of the refrigerator (Fig 5.1a).

Magnets are commonly found in daily life. Let us first study what a magnet is and see what we can learn from it.

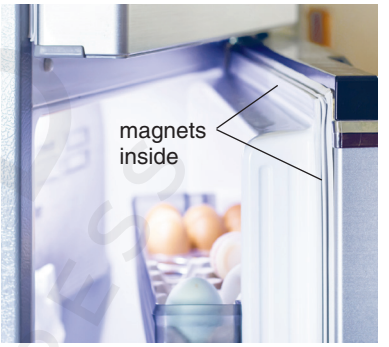


Fig 5.1a A refrigerator door.

1 Magnets

Magnets can attract a few materials, such as iron and steel. These materials are magnetic. They can be *magnetized* and act like magnets. However, most materials like aluminium, copper and wood are unaffected by magnets and are non-magnetic.

A magnet has two poles, namely the north (N) and the south (S) (Fig 5.1b). Both poles can attract magnetic materials.

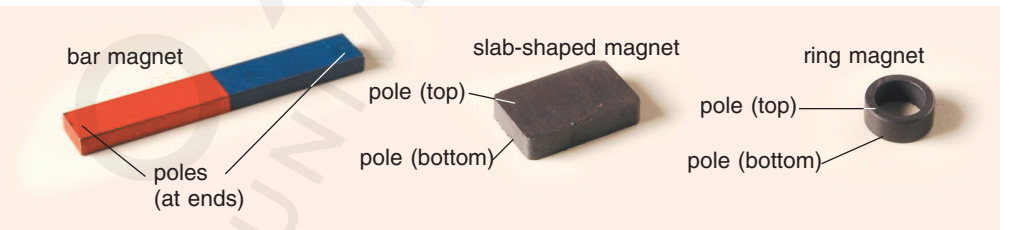


Fig 5.1b Poles of some magnets.

When two like poles are put close to each other, they repel. When two unlike poles are put close to each other, they attract.

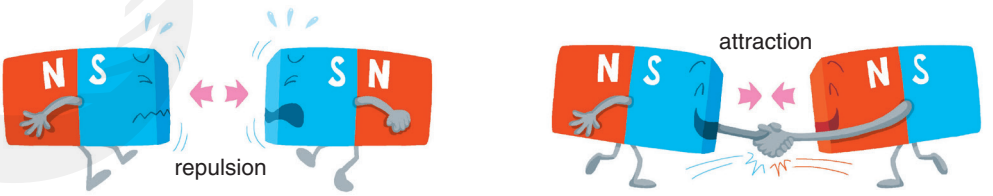


Fig 5.1c Like poles repel; unlike poles attract.

Like poles repel; unlike poles attract.

magnetize 磁化

DSE goal

Examine magnetic field around a magnet.

Describe behaviour of compass in magnetic field.

Represent magnetic field using field lines.

DSE exam

22(1B)Q7

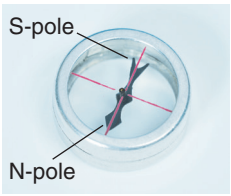
Cross-link

You learned the electric field in Ch 1.3.

Video

→ This video demonstrates Expt 5a.

The needle of a plotting compass is a bar magnet arrow in shape.



- Switch off fans to avoid dispersing iron filings in the air.
- Wear safety goggles to prevent iron filings from entering the eyes.
- Wash your hands thoroughly after the experiment.
- If iron filings enter into the eye accidentally, flush the eye with gently-running water or eye wash unit immediately. Do not rub the eye. Seek medical aid if necessary.

2 Magnetic fields of magnets

A magnet can exert a force on another magnet without touching it. In other words, a **magnetic force** is a non-contact force. We can use the concept of field to study magnetic forces, just like we did in studying electric forces. Let us do Experiment 5a to study the **magnetic fields** set up by magnets.

a Magnetic field patterns

Experiment 5a / Magnetic field patterns

1 Place a Perspex board over a bar magnet.

2 Sprinkle iron filings evenly onto the board (Fig a).

3 Tap the board gently and observe the pattern.

4 Remove the board and the iron filings. Put plotting compasses around the magnet. Observe the pattern shown by the compass needles (Fig b).

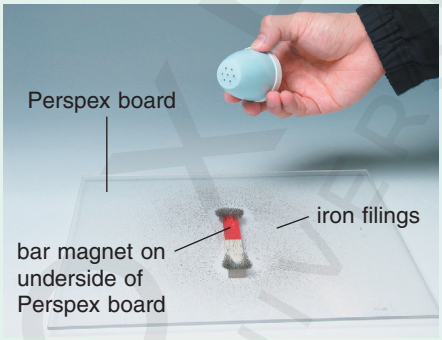


Fig a

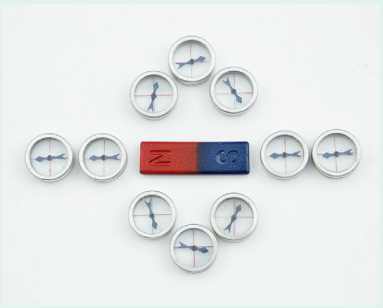


Fig b

5 Repeat steps 1–4 by using

a two bar magnets with unlike poles facing each other,

b two bar magnets with like poles facing each other,

c two slab-shaped magnets with unlike poles facing each other (Fig c).

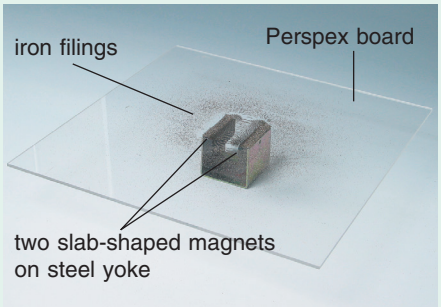


Fig c

Discussion

At which positions is the magnetic field the strongest in each case?

magnetic force 磁力　　magnetic field 磁場

Simulation Video

→ This simulation indicates the magnetic field around a bar magnet by compasses. It is available on OUP web in PC version (.exe) only.

→ This video shows the 3-D magnetic field pattern of a bar magnet.

In Experiment 5a, the iron filings settle into special patterns. Figure 5.1d shows the field pattern of a bar magnet. The field can be represented using **magnetic field lines**.

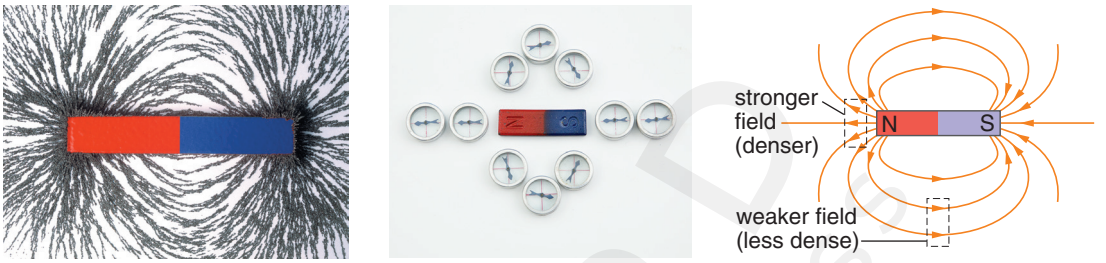


Fig 5.1d *The magnetic field of a bar magnet.*

Note that:

1. Magnetic field lines point from the N-pole to the S-pole.
2. The denser the field lines, the stronger the field (Fig 5.1d). The field is the strongest near the two poles of a magnet.
3. A field line shows the direction of the force acting on the N-pole of another magnet in the field (Fig 5.1e). A **compass needle**, which is a small bar magnet, **aligns with the field line** when it comes to rest.
4. Field lines never cross each other.

The force acting on the S-pole points in the opposite direction of the field line.

In Experiment 5a, you may notice that the iron filings are 'standing' on the Perspex board rather than lying on it.

For simplicity, a field pattern is usually drawn as a two-dimensional figure, but the field is actually three-dimensional (Fig 5.1f).

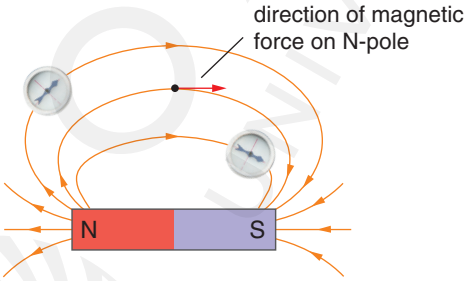


Fig 5.1e *Compass needles align with magnetic field lines.*

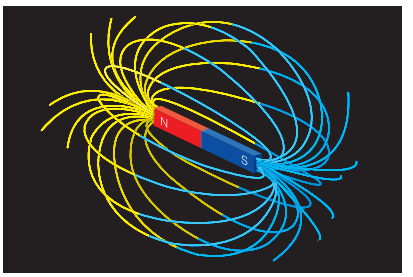


Fig 5.1f *A 3-D magnetic field pattern around a bar magnet.*

Know more

Compasses

How a compass works?
Who invented it?

▸ p.219

Figure 5.1g shows the field pattern between two bar magnets with unlike poles facing each other.



Fig 5.1g *The magnetic field between two bar magnets with unlike poles facing each other.*

magnetic field lines 磁力線

There is a neutral point between two like poles (Fig 5.1h). The field from one magnet cancels the field from the other magnet at this point.

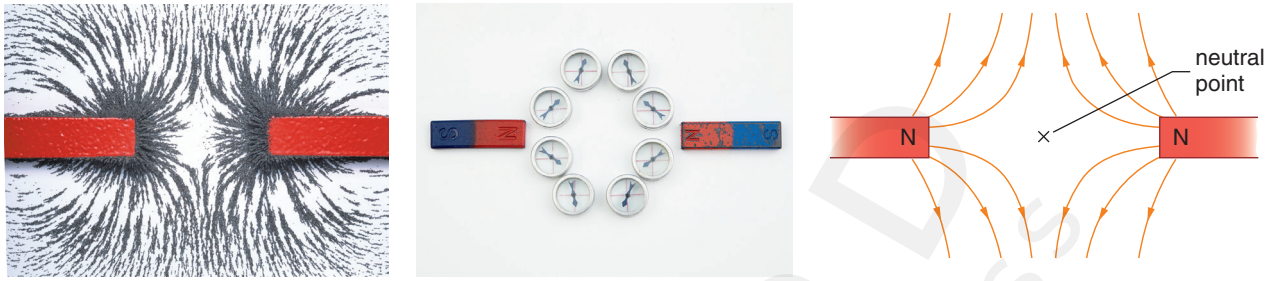


Fig 5.1h *The magnetic field between two bar magnets with like poles facing each other.*

A strong and uniform magnetic field is produced between two slab-shaped magnets with unlike poles facing each other (Fig 5.1i). The uniform field is represented by parallel and evenly-spaced field lines.

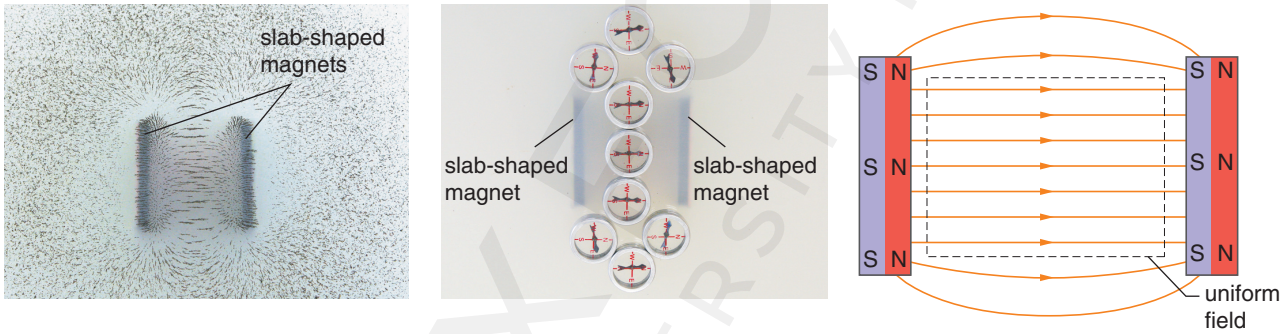


Fig 5.1i *The magnetic field between two slab-shaped magnets with unlike poles facing each other.*

b Earth's magnetic field

The N-pole of a compass needle points to the north. This is because the Earth has a magnetic field pointing approximately from the South Pole to the North Pole (Fig 5.1j). We can imagine that there is a large magnet inside the Earth, with its S-pole close to the North Pole.

The South Pole is a place on the Earth while the south pole is a pole of a magnet.

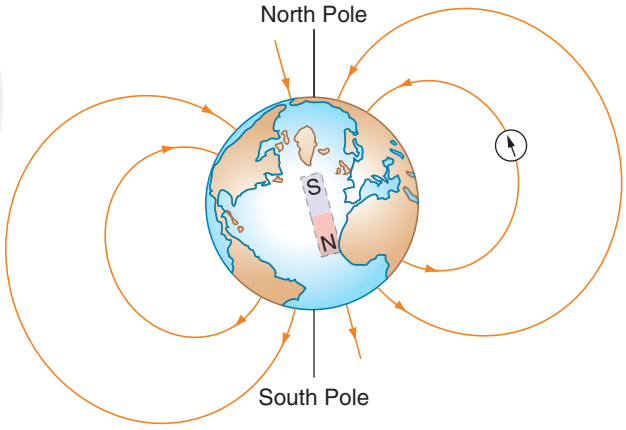


Fig 5.1j *Earth's magnetic field.*

Although 'magnetic field strength' is not exactly the same as 'magnetic flux density', they are interchangeable in secondary level (e.g. HKDSE 2013 Paper 1A Q27, 2017 Paper 1A Q28).

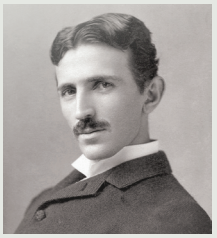
C Measuring a magnetic field

Cross-link
Magnetic flux density and search coil will be further discussed in Ch 7.

The strength of a magnetic field, also know as the *magnetic flux density*, can be measured using a *search coil*, a *hall probe* or a magnetic field sensor (Fig 5.1k). Its unit is the **tesla (T)**.

Know more

Nikola Tesla (1856–1943)



Nikola Tesla was an American inventor. He was best known of designing the a.c. electricity supply system. The unit of magnetic flux density is named after him.

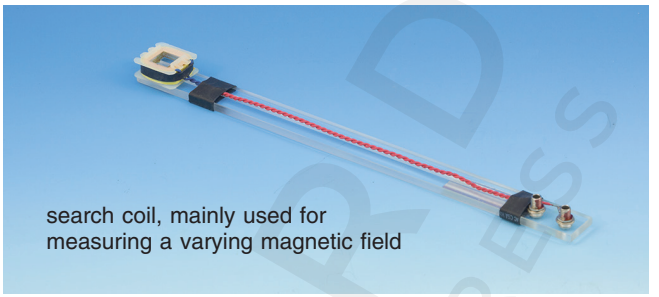


Fig 5.1k *Apparatus for measuring a magnetic field.*

Figure 5.1l shows some typical strengths of magnetic field.

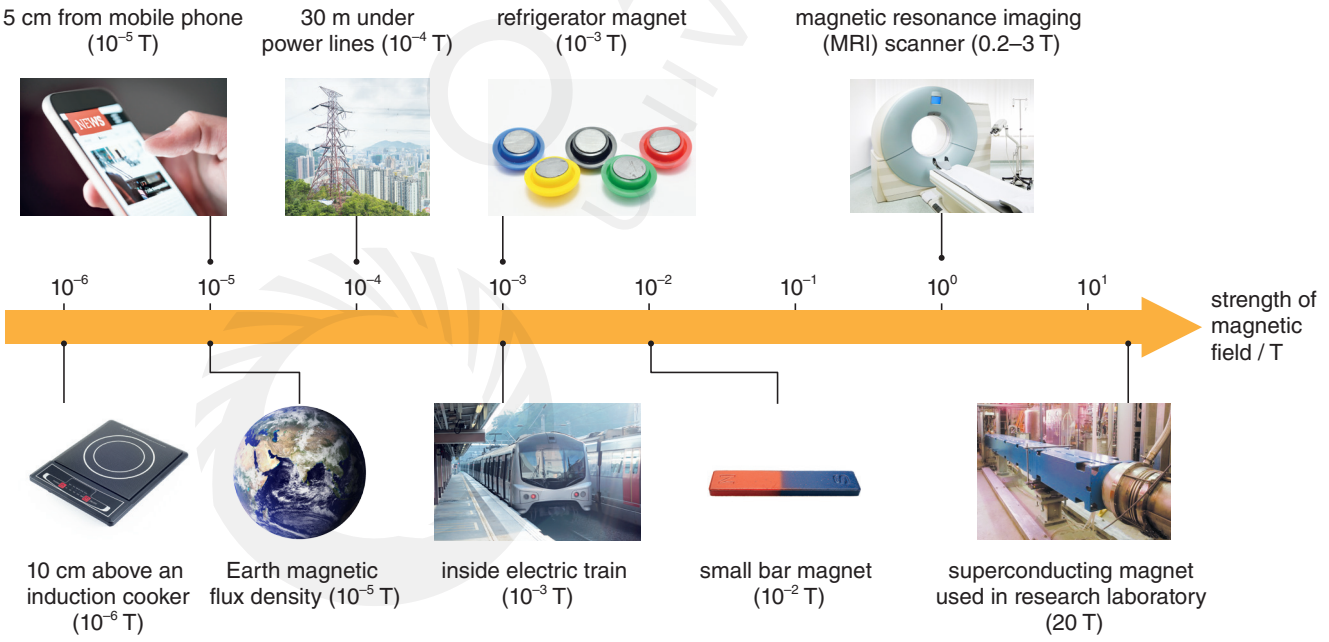


Fig 5.1l *Typical strengths of magnetic field.*

The Earth's magnetic field is very weak. We usually ignore the effect of the Earth's magnetic field unless otherwise stated.

magnetic flux density 磁通量密度　tesla (T) 忒斯拉　search coil 探察線圈
hall probe 霍耳探測器　Nikola Tesla 忒斯拉

Video

→ This video demonstrates the Phone experiment.

Know more

Magnetoreception

Some animals can detect magnetic fields. By detecting the Earth's magnetic field, pigeons can find their way and sea turtles can return to their birth places to lay eggs.



Phone experiment

→ Worksheet is available on OUP web.

Measuring the Earth's magnetic field

The Earth's magnetic field is not parallel to the Earth's surface. It is inclined at different angles at different places.



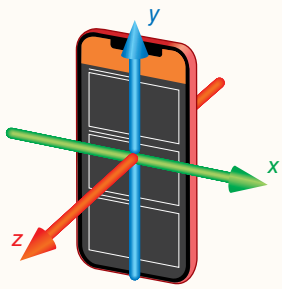
Smartphones are usually equipped with magnetic field sensors. Using a suitable app, you can estimate the strength and direction of the Earth's magnetic field.

What to do

You may search the app on the Internet.

physics phone experiment app

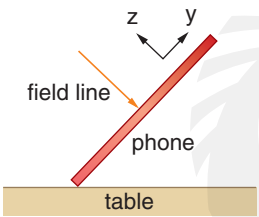
1 Open a *magnetometer* app that allows your phone to detect the magnetic field along the *x*, *y* and *z* directions. The three directions are usually defined as shown.



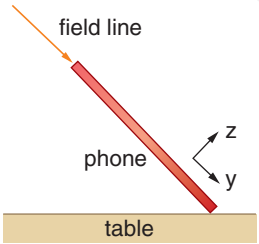
2 Place the smartphone on a table. Make sure there are no magnetic objects (e.g. earphones, quartz watches) or electric appliances nearby.

In step 4, whether the *y* or *z* reading becomes zero depends on how the phone is tilted.

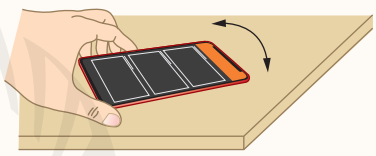
y reading = 0:



z reading = 0:

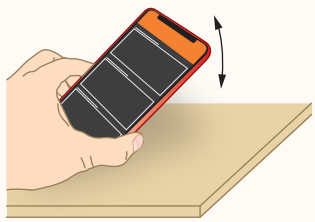


3



Start recording data. The magnetometer should give a lightly fluctuating reading. Rotate the phone on the table until the ***x*** reading becomes approximately zero.

4



Tilt the phone until the ***y*** or ***z*** reading also becomes approximately zero. Now the Earth's magnetic field points along the *z*-axis or the *y*-axis of the phone.

What you find

Does the Earth's magnetic field point upwards or downwards from your table surface?

magnetometer 磁強計

Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

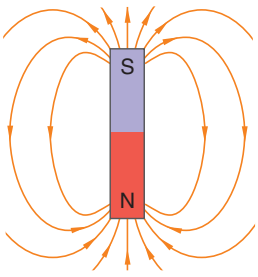
(a) The N-pole of a magnet attracts the N-pole of another magnet. (T/F)

(b) The N-pole of a compass needle is attracted to the South Pole. (T/F)

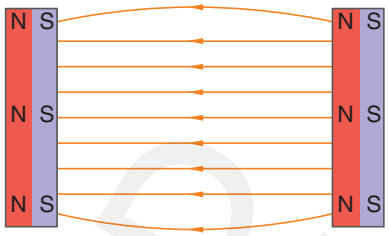
(c) When a magnet is placed at a point in a magnetic field, the force acting on its N-pole points in the same direction as the magnetic field at that place. (T/F)

2 Are the magnetic field patterns below correct?

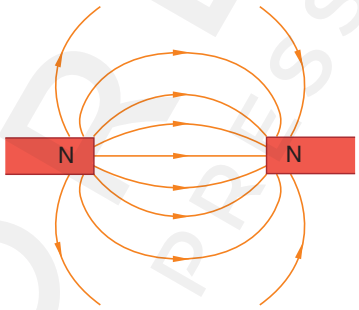
(a)



(b)



(c)



3 The strength of magnetic field is measured in

A N. B T.

C W. D N C^{-1}.

Exercise 5.1

1 Two compass needles point downwards when placed near a bar magnet *X* (Fig a). What is the direction of the needle in compass *P*?

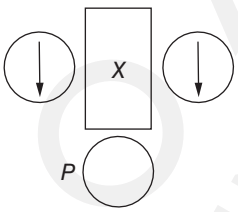


Fig a

A 

B 

C 

D 

★**2** The magnetic fields in the regions between two pairs of magnetic poles *P* and *Q* (Fig b), and *R* and *S* (Fig c) are shown below.

Fig b

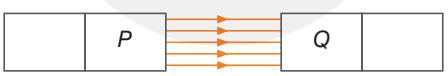


Fig c



Which of the following statements is/are correct?

(1) The magnetic fields are uniform between *P* and *Q*, and between *R* and *S*.

(2) *P* is an N-pole.

(3) The magnetic field between *P* and *Q* is stronger than that between *R* and *S*.

A (1) only B (2) only

C (1) and (3) only D (1), (2) and (3)

★**3** Six bars are wrapped with paper and look alike. Two of them are bar magnets, two are iron bars and two are copper bars. Without using any other tools or unwrapping the bars, how can you identify all of them?

★**4** A bar magnet *X* is placed on a weighing scale. Another bar magnet *Y* is moved towards (without touching) *X* from above (Fig d).

(a) Determine the pole of *Y* facing downwards if the balance reading increases.

(b) If *Y* is an iron bar, how will it affect the balance reading?

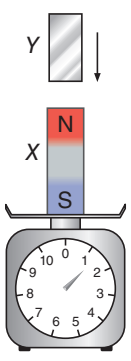


Fig d

5.2 Magnetic effect of electric currents

Video

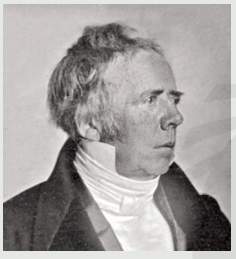
→ This video demonstrates the Hands-on activity.

Do not connect the circuit for a long time. Otherwise, it may overheat.

In 1819, Oersted discovered by chance the magnetic effect of electric current while teaching a class. He noticed that a nearby compass, which happened to be there, deflected when he connected up a circuit for a demonstration.

Know more

Hans C. Oersted (1777–1851)



Oersted was a Danish physicist and chemist. His discovery of an electric current creating a magnetic field was the first connection between electricity and magnetism.

OUP eResources

Hands-on activity

Does an electric current produce a magnetic field?

→ Worksheet is available on OUP web.

What you need



What to do

1



Reshape the paper clip. Connect the terminals of the battery with it to form a closed circuit.

2



Hold the circuit with the glove at different positions close to the compass.

What you find

Why does the compass needle deflect in step 2? The current through the circuit produces a magnetic field.

In the above activity, the compass needle deflects in step 2. This is because a current (or moving charges) can set up a magnetic field (Fig 5.2a). Such an effect was first discovered by *Hans C. Oersted* early in the 19th century.



Fig 5.2a *An electric current sets up a magnetic field and deflects a compass needle.*

Hans C. Oersted 奥斯特

DSE goal

Realize existence of magnetic field due to moving charges or electric current.

Examine magnetic field patterns associated with current through long straight wire, circular coil and long solenoid.

DSE exam

13(1A)Q27, 18(1A)Q27

Video

→ This video demonstrates Expt 5b.

- Switch off fans to avoid dispersing iron filings in the air.
- Wear safety goggles to prevent iron filings from entering the eyes.
- Wash your hands thoroughly after the experiment.
- If iron filings enter into the eye accidentally, flush the eye with gently-running water or eye wash unit immediately. Do not rub the eye. Seek medical aid if necessary.
- Do not touch the hot wire! Switch off the power supply as soon as observations are made to avoid overheating the wire.

1 Magnetic field patterns due to electric currents

We are now going to observe the magnetic field patterns due to electric currents flowing through conductors of various shapes.

Experiment 5b / Magnetic field patterns of electric currents

1 Straight wire

Set up the apparatus to pass a current through the piece of wire (Fig a). Sprinkle iron filings on the board. Tap the board gently and observe the pattern produced.

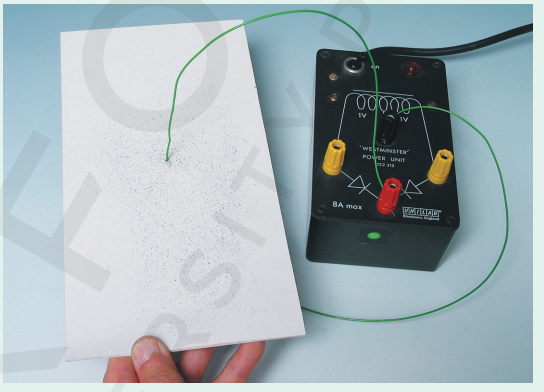


Fig a

2 Flat coil

Repeat by passing a current through a 5-turn flat coil (Fig b). Observe the pattern formed by iron filings.

3 Solenoid

Repeat by passing a current through a 10-turn coil (solenoid) (Fig c). Observe the pattern formed by iron filings.



Fig b

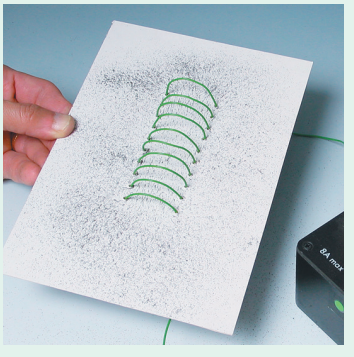


Fig c

Discussion

Suggest a way to find out the direction of the field lines.

✎ The Fleming apparatus is used to show the field lines around a wire. The 'wire' is actually a 10-turn coil of enamel copper wire to enhance the magnetic effect.

We may regard the wires carrying currents in the same direction as a single wire.

a Long straight wires

▶ Figure 5.2b shows the result of step 1 in Experiment 5b. The iron filings form circles around the long straight wire. It shows that the magnetic field is circular with the wire being the centre.

Simulation

→ This simulation shows the magnetic field around a long straight wire. It is available on OUP web in PC version (.exe) only.

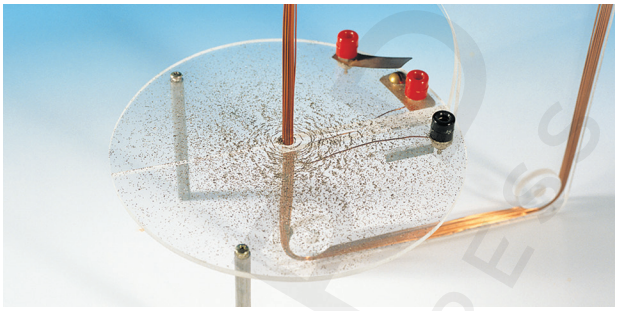


Fig 5.2b Iron filing patterns around current-carrying long straight wires.

✎ May point out that the convention of ⊗ and ⊙ can be used whenever direction is involved, such as for field and force.

We can use plotting compasses to find the direction of the field lines. Figures 5.2c and 5.2d show the results and the corresponding graphical representations (looking from above). The dot ⊙ and the cross ⊗ represent the current flowing out of and into the page respectively. This is like looking at the tip or tail of an arrow.

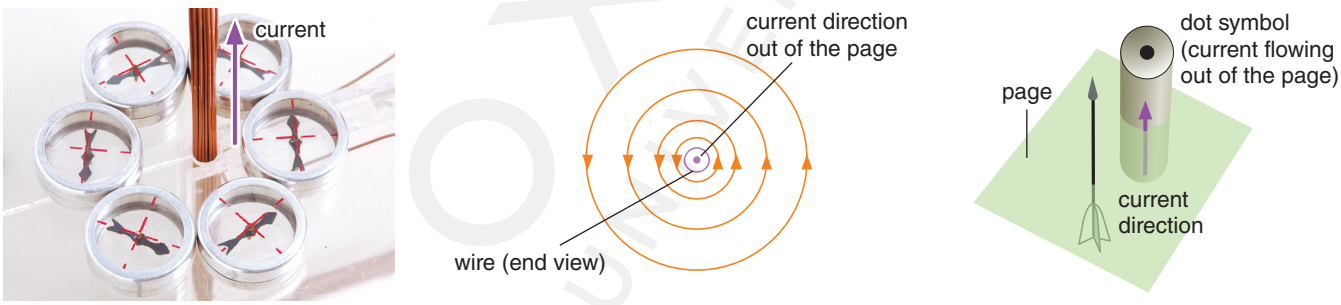


Fig 5.2c Magnetic field pattern around a wire carrying current out of page.

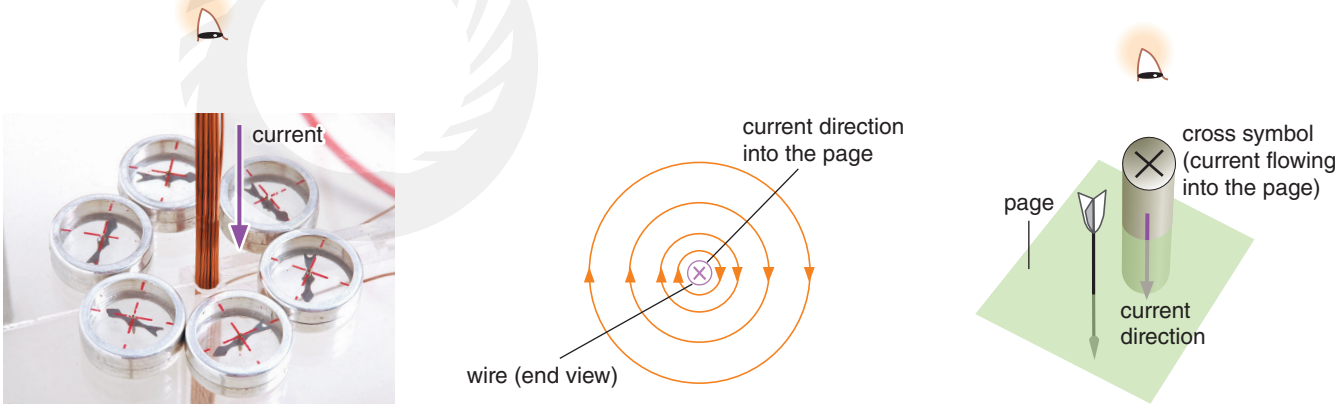


Fig 5.2d Magnetic field pattern around a wire carrying current into page.

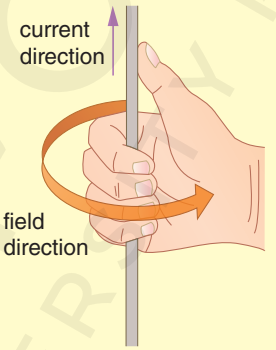
Unlike magnetic field lines, electric field lines never form circles. Otherwise, a charge can accelerate forever along the circle. ▶

The magnetic field pattern around a long straight wire:

1. **The field lines form circles with the wire as the centre.**
2. **Reversing the current reverses the direction of the field lines.**
3. **The field lines are more closely-spaced (the magnetic field is stronger) in the region closer to the wire.**

The direction of the field can be worked out from the direction of the current with the **right-hand grip rule** for long straight wires.

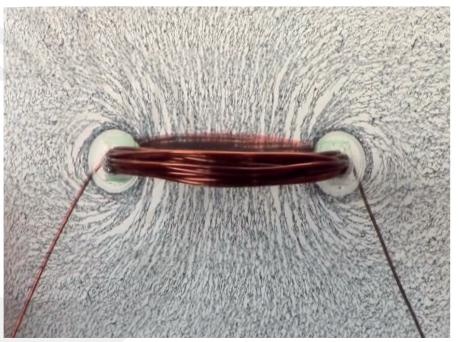
If the right hand grips the wire so that the thumb points the same way as the current, the fingers curl in the same way as the field.



b Circular coils and long solenoids

✎ May show how to use the right-hand grip rule for straight wire to find the direction of magnetic field produced by the circular coil.

Figure 5.2e shows the magnetic field pattern produced by a current flowing through a flat **circular coil**.



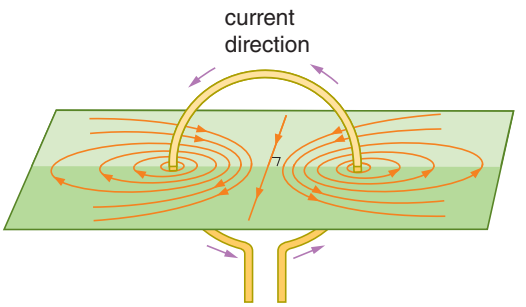


Fig 5.2e *Magnetic field pattern around a current-carrying circular coil.*

At the centre of a circular coil, the field line is straight and at a right angle to the plane of the coil; other field lines run in loops.

right-hand grip rule 右手握拳定則　　circular coil 圓形線圈

A **solenoid** is a long coil made up of many turns of wire. When a current flows through a solenoid, the magnetic field pattern produced is shown in Figure 5.2f.

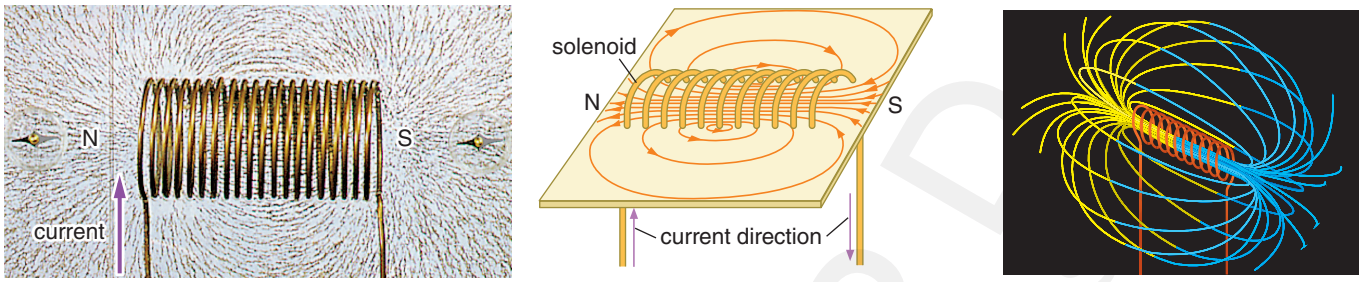
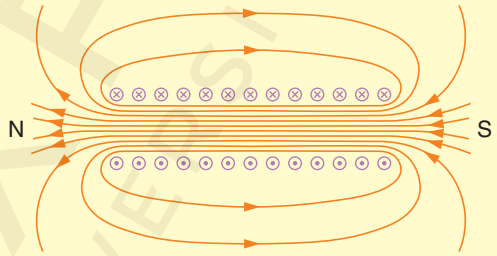


Fig 5.2f *Magnetic field pattern around a current-carrying solenoid.*

1 **Inside a solenoid, the field lines are straight and evenly-spaced, indicating a uniform field.**
2 **Outside a solenoid, the pattern is similar to that around a bar magnet, with one end of the solenoid behaving like an N-pole and the other end like an S-pole.**

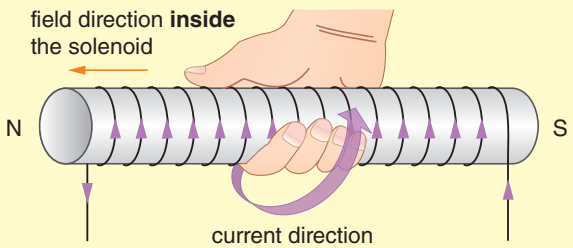


Caution

Pay attention to the difference between the right-hand grip rules for long straight wires and solenoids.

▶ The direction of the magnetic field **inside** a solenoid can be deduced by using the right-hand grip rule for solenoids.

If the right hand grips the solenoid so that the fingers curl in the same way as the current, the thumb points the same way as the field lines inside the solenoid. The end pointed by the thumb is the N-pole.



The direction of the magnetic field of a circular coil can be deduced similarly.

solenoid 螺線管

Example 1 Direction of magnetic field

(a) A long straight wire carries a current into the page (Fig a). Find the direction of magnetic field at *X*.

(b) A long straight wire carries a current upwards (Fig b). Find the direction of magnetic field at *Y*.

(c) A solenoid carries a current as shown (Fig c).

(i) Find the direction of magnetic field inside the solenoid.

(ii) Which end of the solenoid is the S-pole?

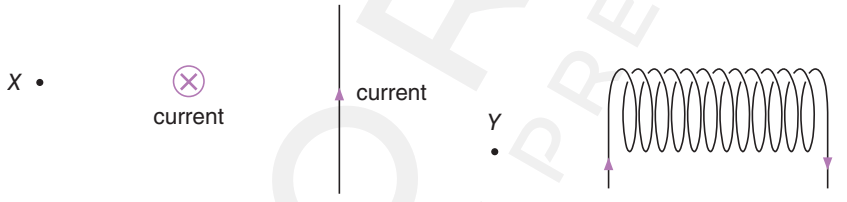


Fig a *Fig b* *Fig c*

Solution

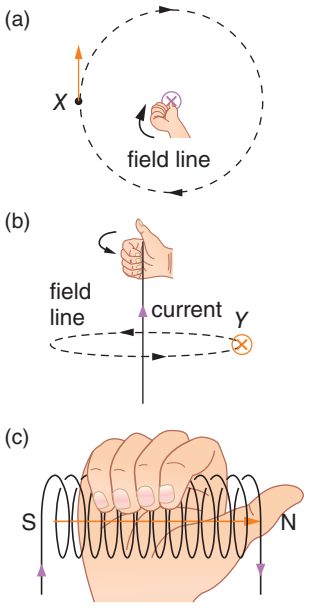
(a) Upwards

(b) Into page

(c) (i) Towards right

(ii) The left end

Tip

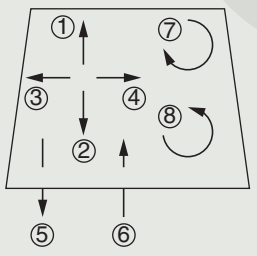


If there are two magnetic fields, the resultant field at a certain point is the vector sum of the two fields at that point. The following is a simple example without calculation.

Know more

Describing direction

We use ① upwards, ② downwards, ③ to the left, ④ to the right, ⑤ into page, ⑥ out of page, ⑦ clockwise and ⑧ anticlockwise to describe directions.



Example 2 Direction of resultant magnetic field

Two parallel straight wires *X* and *Y* carry currents in opposite directions. Find the direction of the resultant magnetic field at point *P*.



Solution

At *P*, both the magnetic fields due to *X* and *Y* point upwards.

∴ The resultant magnetic field at *P* points upwards.

Quick check 2

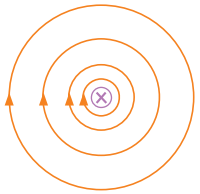
→ E-version in OUP Exercise Platform is also available.

Google Forms

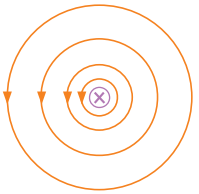
Microsoft Forms

1 A straight wire carries a current into the page. Which of the following figures best shows the magnetic field pattern that it sets up?

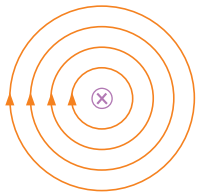
A



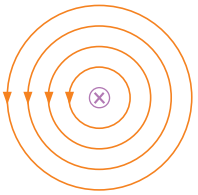
B



C



D



2 Current I passes through a circular coil (Fig a). The magnetic field at P points (into / out of) the page.

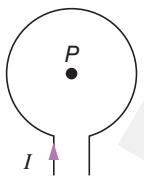


Fig a

3 Current I passes through a solenoid (Fig b).

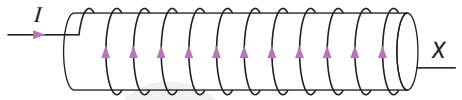


Fig b

(a) The magnetic field inside the solenoid points to the (left / right).

(b) End X is the (N-pole / S-pole).

4 Long straight wires X and Y carry currents in the same direction (Fig c). If wire Y is removed, the magnetic field at point P will (increase / decrease / remain unchanged).

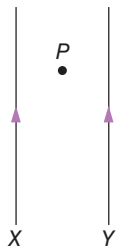
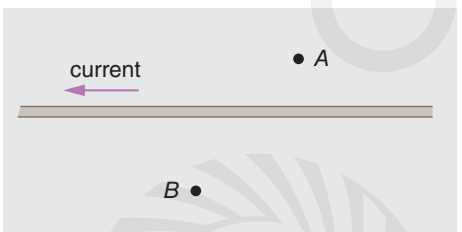


Fig c

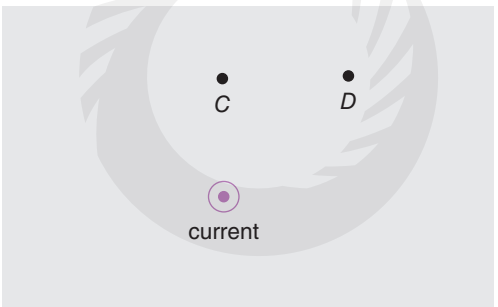
Learn by practice 1

1 In each of the following figures, a current flows in a long straight wire.

(a)



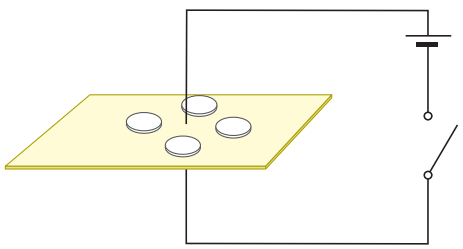
(b)



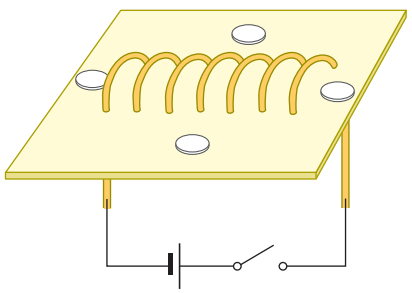
Draw arrows or ⊗ / ⊙ symbols to indicate the directions of the magnetic fields at points A, B, C and D.

2 Draw the directions of the compass needles when the switch is closed in the following cases.

(a)



(b)



DSE goal

Examine factors affecting strength of electromagnet.

DSE exam

20(1B)Q3

2 Electromagnets

A solenoid behaves like a bar magnet when there is current passing through it. So, it can be used as an **electromagnet**. It is a magnet that can be turned on and off easily (Fig 5.2g).

(i) 

(ii) 

Fig 5.2g *An electromagnet that is (i) turned on, (ii) turned off.*

a Strength of an electromagnet

Video

→ This video demonstrates Expt 5c.

Experiment 5c Factors affecting the strength of an electromagnet

1 Insert a soft-iron core in a plastic tube (Fig a). Use an insulated copper wire to wind a coil of about 50 turns onto the tube. Connect the coil to the power supply to make an electromagnet. A rheostat and an ammeter are also connected in series to the coil.

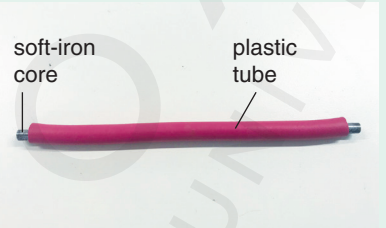


Fig a

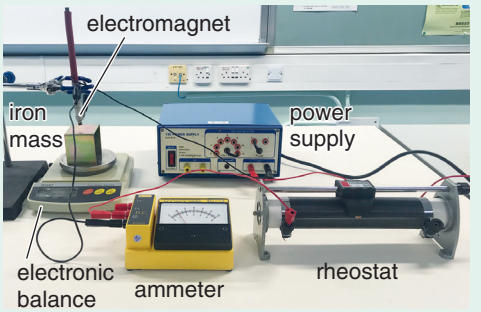


Fig b

2 Put an iron mass on an electronic balance and set the reading to zero.

3 Fix the electromagnet above the mass and turn on the power supply (Fig b). Record the balance readings with

(a) different current passing the electromagnet,

(b) electromagnets wound with different number of turns within the same length and

(c) electromagnets with cores made of different materials.

Discussion

1 Why is the balance reading always negative when current passes the coil?

2 State the factors that affect the strength of an electromagnet.

✎ May use the number of paper clips that the electromagnet can attract to test its strength.

✎ Soft-iron and steel contain different proportions of iron and hence have different hardness. Soft-iron is so named because it is softer than steel. Steel core can retain magnetism after the current is cut off while soft-iron core cannot. Therefore, magnetic materials which can retain magnetism are called 'hard'; those failing to retain magnetism are called 'soft'.

electromagnet 電磁鐵

From the experiment,

an electromagnet can be made stronger by
1 increasing the number of turns per unit length of the coil,
2 increasing the current through the coil,
3 inserting a soft-iron core into the coil.

An electromagnet powered by a dry cell is strong enough to hold a man of weight 600 N by making good use of these 3 factors.



Note that a core made of other **magnetic materials** can also strengthen the magnetic field.

Example 3 Electromagnets

A student connects a solenoid to a battery to make an electromagnet (Fig a).

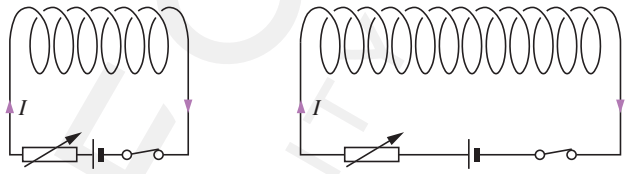


Fig a *Fig b*

(a) How does the strength of the electromagnet change when the student reduces the resistance of the variable resistor?

(b) Suggest an advantage of an electromagnet over a permanent magnet.

(c) Both a soft-iron core and a steel core can strengthen the electromagnet. However, after the switch is opened, the steel core remains magnetized (i.e. behaves like a magnet) but the soft-iron core does not. Which core is more suitable in making an electromagnet? Explain.

(d) The student connects another solenoid in series to the original one (Fig b). The two solenoids are identical and their resistances are negligible. How does the strength of the electromagnet change? Explain.

Solution

(a) When the resistance decreases, the current through the solenoid increases. Therefore, the strength of the electromagnet increases.

(b) It can be turned off. / Its strength can be varied.

(c) The soft-iron core is more suitable. If the steel core is used, the electromagnet cannot be turned off.

(d) The strength remains unchanged because the number of turns per unit length of the solenoid remains unchanged.

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 A coil is connected to a power supply via a resistor (Fig a).

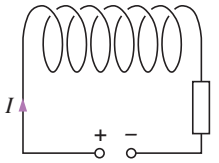


Fig a

Which of the following methods can increase the magnetic field produced? Put a tick in the corresponding box(es).

- ☐ Increase the voltage output of the power supply.
- ☐ Compress the coil to reduce its length.
- ☐ Insert a copper core into the coil.
- ☐ Replace the resistor with another one with a higher resistance.

2

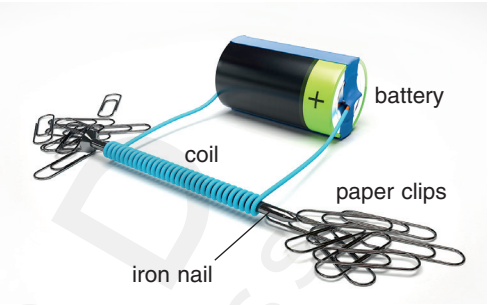


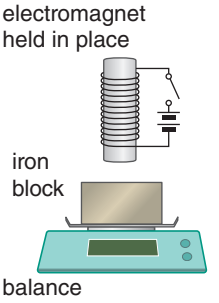
Fig b

Which of the following about the electromagnet shown in Figure b is correct? Put a tick in the corresponding box(es).

- ☐ If the coil is disconnected from the battery, the paper clips will lose the attraction immediately.
- ☐ If the polarity of the battery is reversed, the paper clips will be repelled away.
- ☐ If the iron nail is removed, the paper clips will lose the attraction immediately.

Learn by practice 2

1 An iron block is put on a balance and the reading of the balance is set to zero. An electromagnet is fixed above the iron block.

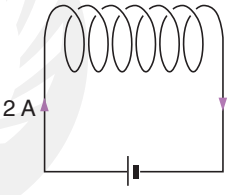


(a) How does the balance reading change when the electromagnet is turned on?

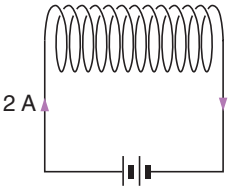
(b) Suggest two methods that can make the change in the balance reading more significant when the electromagnet is turned on.

2 Arrange the following set-ups in ascending order of the strength of the magnetic field produced.

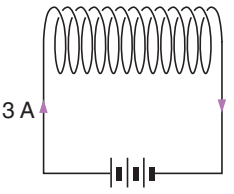
I



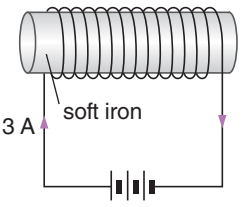
II



III



IV



The lifting capacity of a heavy-duty electromagnetic crane can be up to 2×10^5 kg. The electromagnet crane cannot completely replace other mechanical lifting devices because it is unable to lift nonmagnetic objects.

b Applications of electromagnets

Electromagnets have many applications. Figure 5.2h shows some of them.

(i) An electromagnetic crane *(ii) A pair of earphones* *(iii) A maglev train*



Fig 5.2h *Some applications of electromagnets.*

Know more

Applications of electromagnets

Learn more about the working principle of earphones and maglev trains.

▸ p.220

Example 4 Electric bell

An electric bell (Fig a) makes use of an electromagnet. Figure b shows the schematic diagram of an electric bell. Explain how it works.



Fig a

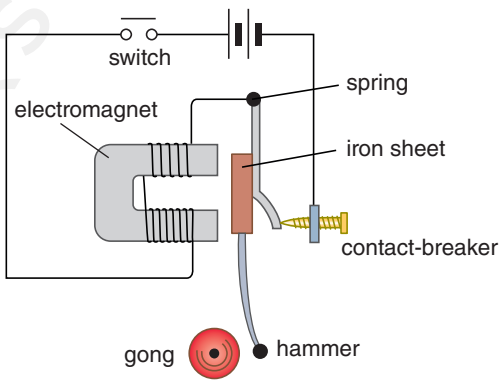


Fig b

Solution

When the switch is pressed, the circuit becomes closed. The iron sheet attached to the hammer is attracted by the electromagnet, so the hammer hits the gong. This breaks the connection and the circuit becomes open (Fig c). As a result, the electromagnet loses its magnetic effect. The hammer springs back until it touches the contact-breaker. The circuit is closed again and the whole process is repeated as long as the switch is pressed.

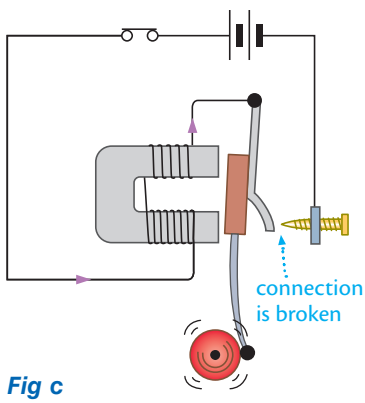


Fig c

DSE goal

Apply $B = \frac{\mu_0 I}{2\pi r}$ and $B = \frac{\mu_0 NI}{l}$ to represent magnetic fields around a long straight wire and inside a long solenoid, and solve related problems.

DSE exam

14(1A)Q26, 28, 21(1A)Q25

3 Calculating magnetic fields

In this part, we will learn how to calculate the strength of the magnetic field produced by a current-carrying conductor.

a Long straight wires

Figure 5.2i shows a current flowing through a long straight wire.

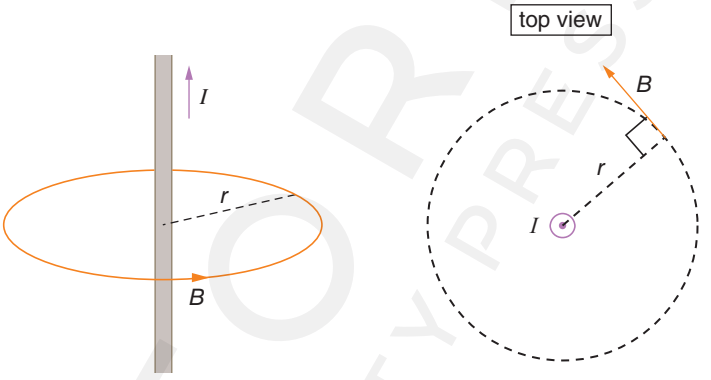


Fig 5.2i *Magnetic field produced by a current through a long straight wire.*

The magnetic field B is circular with the wire as the centre. At a distance r from the wire, the strength of B is given by:

This equation gives the theoretical value of B around an infinitely long wire. When the length of the wire is much larger than r, this equation provides a good approximation to B.

permeability of free space = $4\pi \times 10^{-7}$ T m A^{-1}

magnetic field (T)

$$B = \frac{\mu_0 I}{2\pi r}$$

current (A)

distance from wire (m)

Different materials have different permeability.

where μ_0 is a constant known as the *permeability* of free space and has the exact value of $4\pi \times 10^{-7}$ T m A^{-1}. Note that:

1. For a fixed r, B is directly proportional to I ($B \propto I$). A larger current sets up a stronger magnetic field (Fig 5.2j).
2. For a fixed I, B is inversely proportional to r $\left(B \propto \frac{1}{r}\right)$. The magnetic field becomes weaker when the distance from the wire increases (Fig 5.2k).

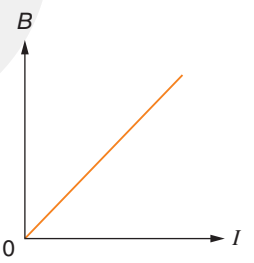


Fig 5.2j *Graph of B against I (constant r) of a long straight wire.*

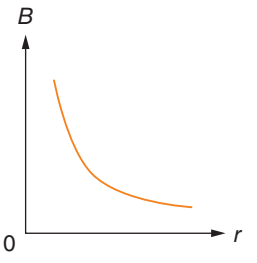


Fig 5.2k *Graph of B against r (constant I) of a long straight wire.*

permeability 磁導率

Caution

Do not mix up the prefix μ (10^{-6}) with the constant μ_0.

Example 5 Distance from wire

A current of 0.5 A passes through a long straight wire. Point P is at a distance d from the wire and the magnetic field there has a strength of 4 μT. Find d. Take $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}.

Solution

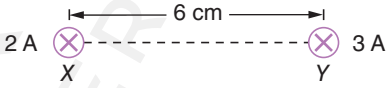
By $B = \dfrac{\mu_0 I}{2\pi r}$,

$$4 \times 10^{-6} = \frac{(4\pi \times 10^{-7}) \times (0.5)}{2\pi(d)}$$

$$d = 0.025 \text{ m}$$

Example 6 The neutral point between two wires

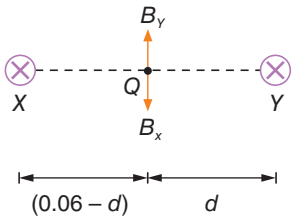
Two parallel long wires X and Y carry currents of 2 A and 3 A respectively into the page. They are 6 cm apart.



Q is a neutral point between wires X and Y. Find the distance d of Q from wire Y.

Solution

At Q, the magnetic field due to X (B_X) points downwards and the magnetic field due to Y (B_Y) points upwards. They cancel each other out.



Therefore, by $B = \dfrac{\mu_0 I}{2\pi r}$,

$$B_X = B_Y$$

$$\frac{\mu_0(2)}{2\pi(0.06 - d)} = \frac{\mu_0(3)}{2\pi(d)}$$

$$2d = 3(0.06 - d)$$

$$d = 0.036 \text{ m}$$

Think about...

If the current in X increases, how will the position of the neutral point change?

Know more

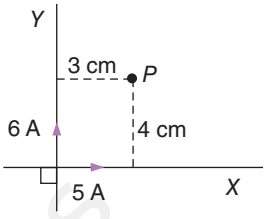
Coaxial cable

A coaxial cable is a specially designed cable which produces no magnetic field outside the cable. Do you know how it works?

▶ p.219

Example 7 | Calculating resultant magnetic field

Two long straight wires X and Y are perpendicular to each other. They carry currents of 5 A and 6 A as shown. Take $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}.



(a) Find the resultant magnetic field at point P.

(b) The measured value of the magnetic field at P is smaller than the answer in (a). Suggest a reason.

Solution

(a) Magnetic field at P due to $X = \dfrac{\mu_0 I}{2\pi r} = \dfrac{(4\pi \times 10^{-7})(5)}{2\pi(0.04)} = 2.5 \times 10^{-5}$ T (out of page)

Magnetic field at P due to $Y = \dfrac{\mu_0 I}{2\pi r} = \dfrac{(4\pi \times 10^{-7})(6)}{2\pi(0.03)} = 4 \times 10^{-5}$ T (into page)

$\therefore$ Resultant magnetic field at $P = 4 \times 10^{-5} - 2.5 \times 10^{-5} = 1.5 \times 10^{-5}$ T (into page)

(b) The Earth's magnetic field at P points out of the page.

Quick check 4

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

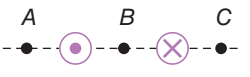
Take $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}.

1 Current I flows through a long straight wire. Complete the table.

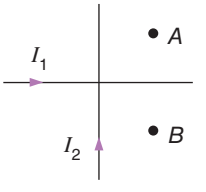
	B / T	I / A	r / m
(a)		2	0.05
(b)	3×10^{-6}		0.1

2 Each figure below shows two long straight wires carrying currents. At which point(s) may the resultant magnetic field be zero?

(a)



(b)



Learn by practice 3

(a) P: 6×10^{-6} T (out of page)
Q: 4×10^{-6} T (into page)

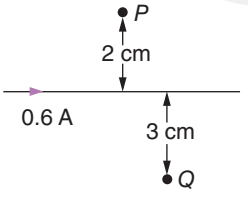
(b) P: 3×10^{-5} T (to the left)
Q: 1.5×10^{-5} T (downwards)

(c) P: 8.86×10^{-6} T (out of page)
Q: 6×10^{-6} T (out of page)

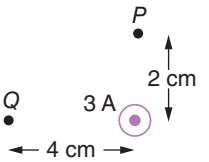
(d) P: 2.33×10^{-6} T (into page)
Q: 7.67×10^{-6} T (out of page)

1 Find the strengths and directions of the magnetic fields at points P and Q in each case. Take $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}.

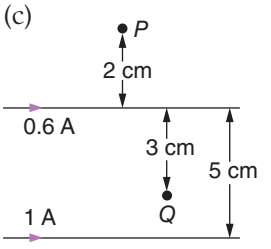
(a)



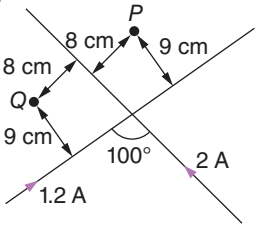
(b)



(c)



(d)



You can find the position of the magnetic field sensor in the phone like this: Move a small magnet to different positions above the phone. When the phone shows the maximum reading, the sensor is right below the magnet.

Do not use side bench for this experiment as there may be many cables around it. Use a long wire of 2–3 m and stick it on a long bench.

Example 8 Estimating the permeability of free space

Dick connects a long straight wire as shown (Fig a). He puts a smartphone with a magnetic-field-detector app at a distance r from the wire and measures the magnetic field perpendicular to the page $B_\perp$. Then he varies r and measures the corresponding values of $B_\perp$. Figure b shows his results. The ammeter reading is 5 A.

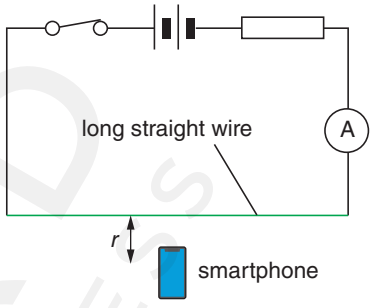


Fig a

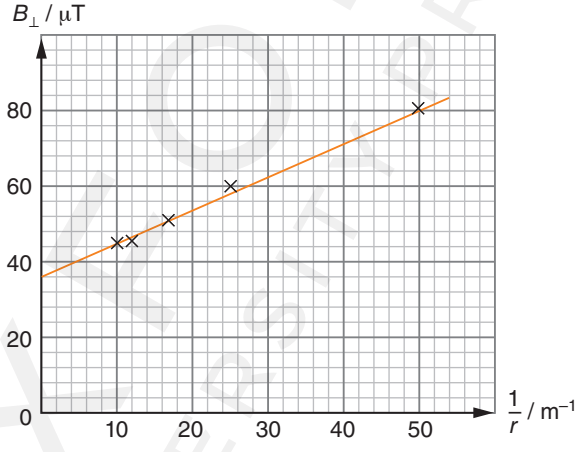


Fig b

(a) Why does the graph not pass through the origin?

(b) Estimate the value of μ_0 from the graph.

Solution

(a) The Earth's magnetic field B_E contributes to the value of $B_\perp$.

(b) The magnetic field produced by the current $B_C = \dfrac{\mu_0 I}{2\pi r}$

$$B_\perp = B_C + B_E$$

$$= \frac{\mu_0 I}{2\pi} \times \frac{1}{r} + B_E$$

$$\therefore \text{ Slope of graph} = \frac{\mu_0 I}{2\pi}$$

$$\frac{(80 - 36) \times 10^{-6}}{50 - 0} = \frac{\mu_0 (5)}{2\pi}$$

$$\mu_0 = 1.11 \times 10^{-6} \text{ T m A}^{-1}$$

Revision 5 Q4 (p.213)

b Inside a long solenoid

Figure 5.2l shows a long solenoid of length l and radius r with N turns. A current I passes through it.

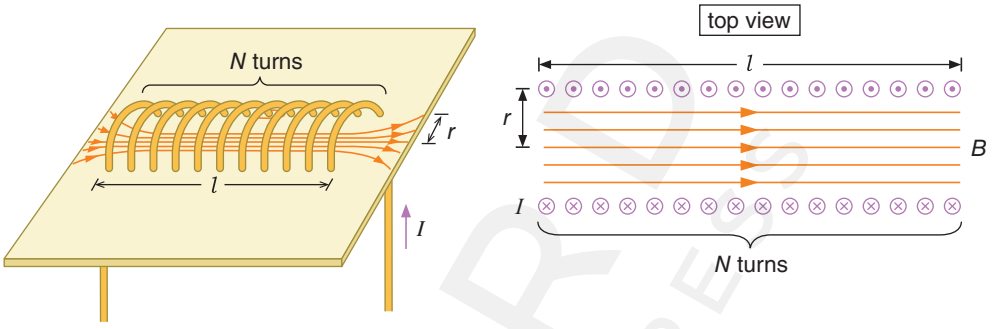
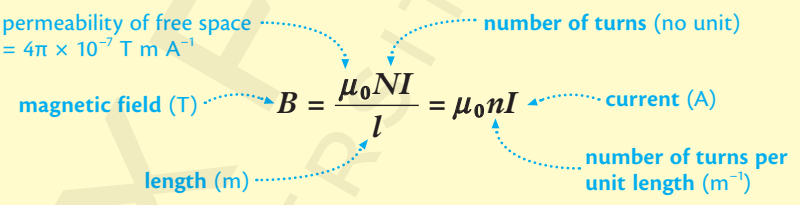


Fig 5.2l *Magnetic field inside a long solenoid.*

The uniform magnitude field inside the long solenoid (far away from both ends) has a strength

This equation gives a good approximation to a solenoid with $l >> r$. ▶



where $n = \frac{N}{l}$. Note that:

1 For a fixed n, B is directly proportional to I ($B \propto I$). A larger current produces a stronger magnetic field inside the solenoid (Fig 5.2m).

2 For a fixed I, B is directly proportional to n ($B \propto n$). The magnetic field inside the solenoid becomes stronger when the separation of the turns of the coils decreases (Fig 5.2n).

3 The magnetic field inside a solenoid is independent of r, if l is much larger than r.

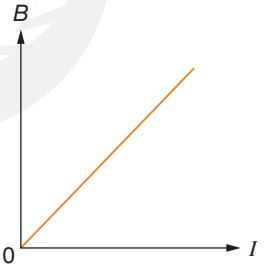


Fig 5.2m *Graph of B against I (constant n) of a long solenoid.*

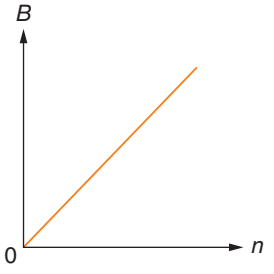


Fig 5.2n *Graph of B against n (constant I) of a long solenoid.*

Example 9 Magnetic field inside a long solenoid

A current of 2 A is flowing through a long solenoid having a length of 25 cm and 200 turns. Take $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}.

(a) Find the strength of the magnetic field near the centre inside the solenoid.

(b) Explain how the answer in (a) is affected if

(i) the length and the number of turns are both doubled;

(ii) the cross-sectional area is reduced by half,

(iii) the current through the solenoid is doubled.

Solution

(a) $B = \dfrac{\mu_0 NI}{l}$

$= \dfrac{(4\pi \times 10^{-7}) \times 200 \times 2}{0.25}$

$= 2.01 \times 10^{-3}$ T

(b) (i) There is no change as the number of turns per unit length n remains the same.

(ii) There is no change as the magnetic field inside a long solenoid is independent of the cross-sectional area.

(iii) Since $B \propto I$,

B will be doubled when I is doubled.

Google Forms Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 The magnetic field at the centre of a solenoid is doubled when

☐ the current is doubled.

☐ the length of the solenoid is doubled (by connecting an identical solenoid in series).

☐ the separation of the turns is doubled.

☐ the diameter of the solenoid is doubled.

2 Complete the following table about the magnetic field inside a long solenoid. Take $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}.

	N	l / m	I / A	B / μT
(a)	100	1	1	
(b)	200	2		50
(c)	20		2	100

Exercise 5.2

Take $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}.

★**1** A bar magnet is placed on a weighing scale and an electromagnet is fixed right above it (Fig a). What happens to the reading of the scale when the switch is closed?

A It increases.

B It decreases.

C It does not change.

D Cannot be determined.

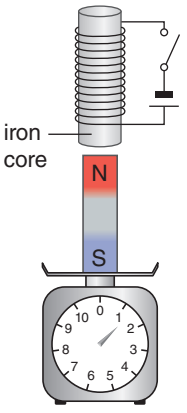


Fig a

★**2**

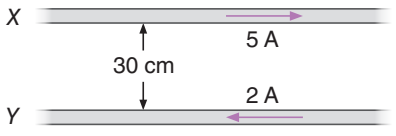


Fig b

Figure b shows two long straight parallel wires X and Y carrying currents of 5 A and 2 A respectively. The two wires are 30 cm apart. P is a neutral point in the magnetic field due to wires X and Y. What is the distance of P from wire Y?

A 14 cm

B 20 cm

C 42 cm

D 50 cm

★**3** A solenoid has N turns and its length is L. When a current I passes through it, the magnetic field inside it is B_0. What is the magnetic field inside it if it is stretched to a length $2L$ and the current passing through it becomes $2I$?

A $\frac{B_0}{4}$ B B_0

C $2B_0$ D $4B_0$

★**4** A long solenoid has 10 turns of coil in every 1.5 cm and carries a current of 5 A (Fig c).

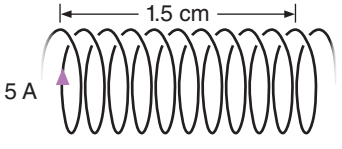


Fig c

(a) Find the strength and direction of the magnetic field at the centre of the solenoid.

(b) Which end of the solenoid is the N-pole?

(c) How will the answer of (a) change if the solenoid (i) is compressed; (ii) has a larger radius?

★**5** Figure d shows a horseshoe electromagnet.

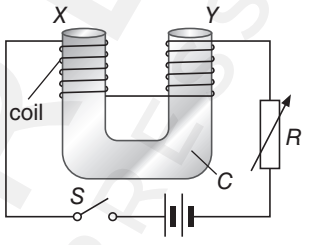


Fig d

(a) Component C makes the electromagnet stronger when S is closed (but retains no magnetism when S is opened). Suggest a material that C is made of.

(b) Determine what the poles at X and Y are when S is closed.

(c) Explain how the strength of the electromagnet will be affected if

(i) the resistance of R increases;

(ii) the e.m.f. of the battery increases;

(iii) the coil is made of thicker wires.

★**6** David sets up the following apparatus using springs and a coil wound around a movable frame (Fig e). He claims that the set-up can be used to measure the size of a current.

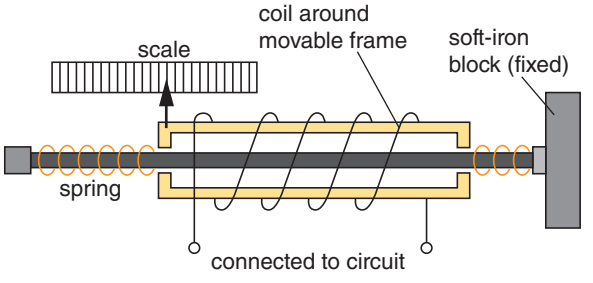


Fig e

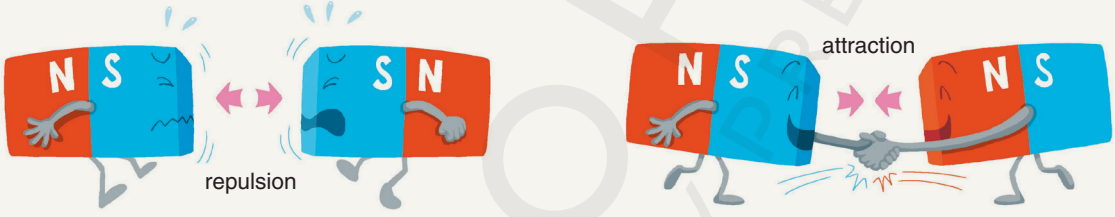
(a) Explain how the apparatus works.

(b) How can the apparatus be modified so that it can also measure the direction of the current?

Key concepts 5

1 Magnets

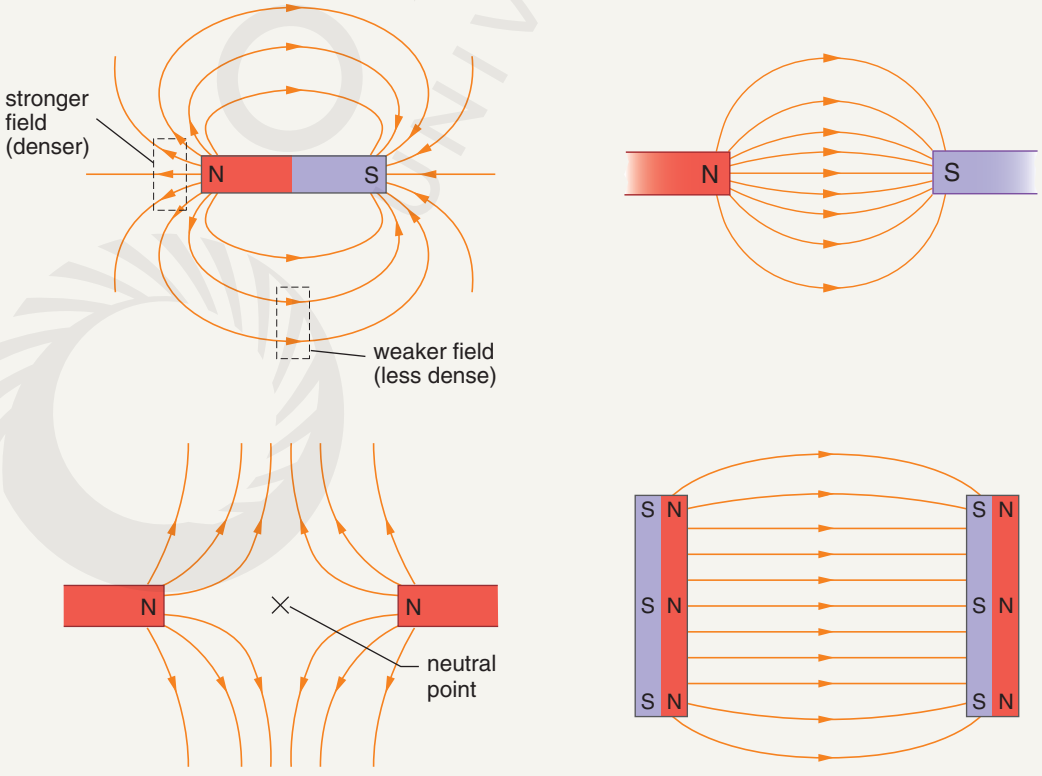
- magnets attract magnetic materials, e.g. iron
- magnets exert force on one another:
  - like poles repel each other
  - unlike poles attract each other



2 Magnetic fields

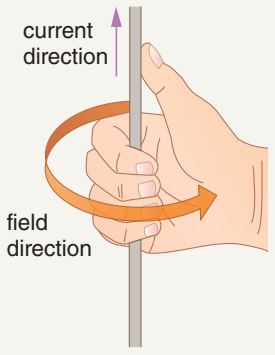
Magnetic fields produced by magnets

- magnetic field lines point from the N-pole to the S-pole outside a magnet.

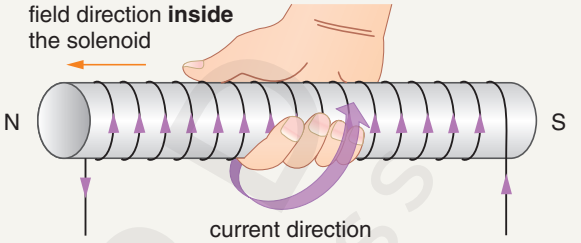


Methods for determining the directions of magnetic field produced by currents

right-hand grip rule for straight wires

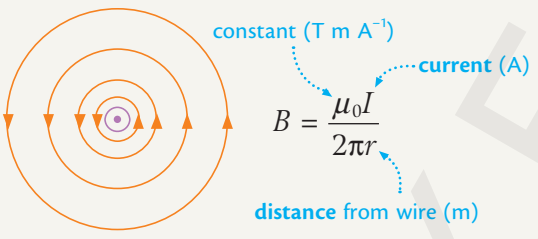


right-hand grip rule for solenoids

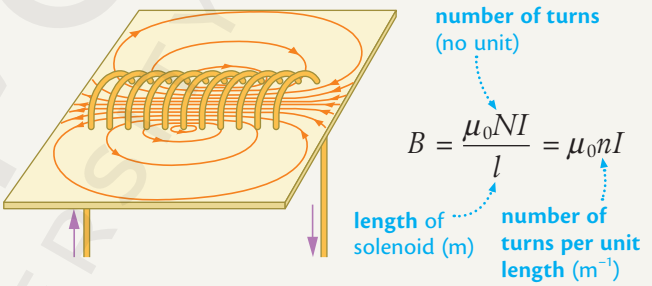


Magnetic fields *B* produced by currents

around a long straight wire

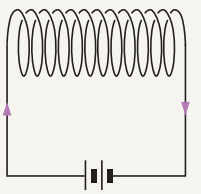


inside a long solenoid



3 Electromagnets

- a current-carrying coil, usually with a core
- can be strengthened by
  - increasing the number of turns per unit length of the coil
  - increasing the current through the coil
  - inserting a soft-iron core into the coil



Key terms

1	circular coil 圓形線圈	p.196	**5**	magnetic force 磁力	p.187
2	electromagnet 電磁鐵	p.200	**6**	right-hand grip rule 右手握拳定則	p.196
3	magnetic field 磁場	p.187	**7**	solenoid 螺線管	p.197
4	magnetic field lines 磁力線	p.188	**8**	tesla (T) 忒斯拉	p.190

Revision 5

Unless otherwise specified, the effect of the Earth's magnetic field can be ignored.

Take $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}.

Multiple-choice questions

★1
5.2

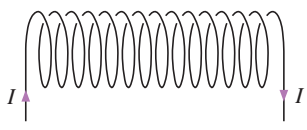


Fig a

Which of the following actions can cause the long solenoid above to give a stronger magnetic field inside the solenoid?

(1) Insert a copper core into the solenoid.

(2) Compress the solenoid.

(3) Stretch the solenoid.

A (2) only

B (3) only

C (1) and (2) only

D (1) and (3) only

★2
5.2

Two electromagnets are placed as shown below (Fig b). *X* and *Y* are identical solenoids.

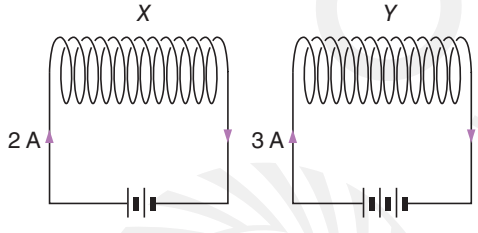


Fig b

Which of the following statements are correct?

(1) *Y* produces a stronger magnetic field than *X*.

(2) The magnetic force exerted on *X* by *Y* is stronger than the magnetic force exerted on *Y* by *X*.

(3) *X* and *Y* attract each other.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★3
5.2

A current of 3 A flows through a long solenoid *X* of 200 turns. A current of 5 A flows through a long solenoid *Y* of 300 turns. The cross-sectional area of *X* is 25 cm^2 while that of *Y* is 50 cm^2. *X* and *Y* have the same length. What is the ratio of the strength of the magnetic field inside solenoid *X* to that inside solenoid *Y*?

A 2 : 5

B 3 : 5

C 2 : 3

D 9 : 10

★4
5.2

A student passes a constant current *I* through a long straight wire and uses a magnetic field sensor to measure the magnetic field *B* produced by the current at different distances *d* from the wire. The result is shown in Figure c.

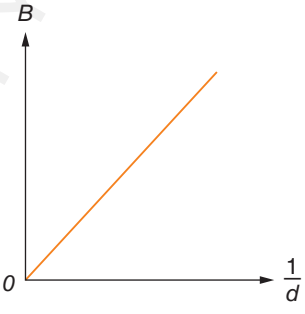
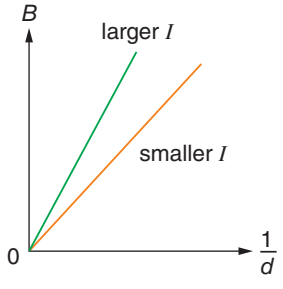


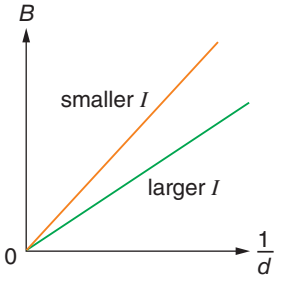
Fig c

If the student repeats the experiment with a larger current, which of the following figures best shows the new graph (line in green)?

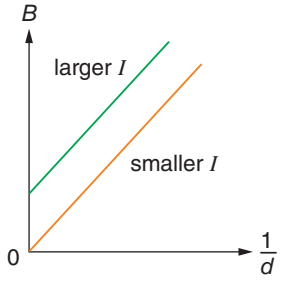
A



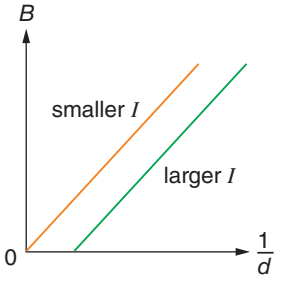
B



C



D



★5
5.2

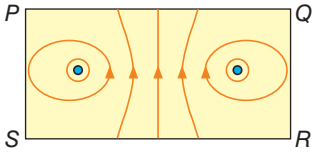
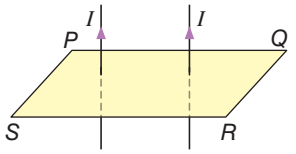


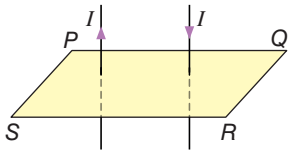
Fig d

Which of the following settings can produce the resultant magnetic field pattern shown above?

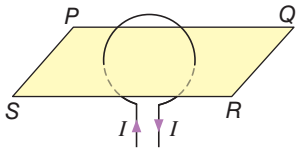
(1)



(2)



(3)



A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

★6
5.2

A bar magnet is placed on a weighing scale, and an electromagnet is right above it (Fig e). Which of the following actions will increase the reading of the scale?

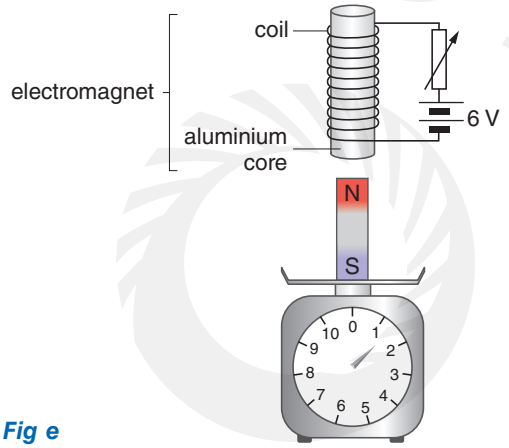


Fig e

A Move the electromagnet slightly downwards.
B Take away the aluminium core.
C Reduce the resistance of the variable resistor.
D Halve the number of turns per unit length of the coil.

★7
5.2

X and Y are two parallel long straight wires carrying currents of 1 A and 2 A respectively (Fig f). The strength of the resultant magnetic field at point P is B_P. What is the strength of the resultant magnetic field at point Q?

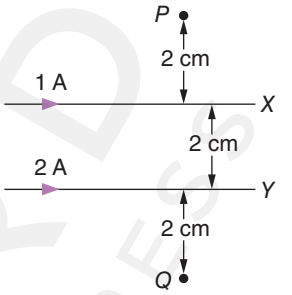


Fig f

A $\frac{2}{3}B_P$
B $\frac{4}{5}B_P$
C $\frac{5}{4}B_P$
D $\frac{3}{2}B_P$

★8
5.2

P, Q, R and S are four parallel long straight wires located at the corners of a regular hexagon (Fig g). All wires carry the same magnitude of current in the directions shown. O is the centre of the hexagon. Should the current in which wire be reversed so that the resultant magnetic field at O become zero?

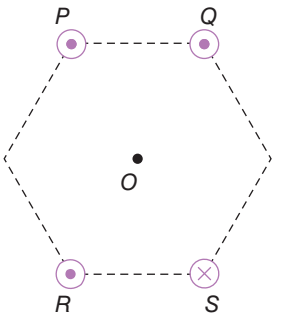


Fig g

A P only
B Q only
C R only
D None of the above.

***9** 5.2 P and Q are two parallel long straight wires (Fig h). The current I in Q flows downwards. Point X is $2d$ from P and d from Q. If the resultant magnetic field at X is zero, what are the magnitude and direction of the current in P?

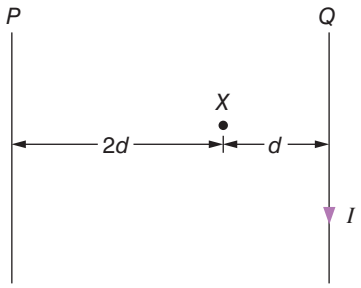


Fig h

A $\frac{I}{2}$ (upwards)

B $\frac{I}{2}$ (downwards)

C $2I$ (upwards)

D $2I$ (downwards)

****10** 5.2 P, Q, R, S, T and U are six parallel long straight wires located at the corners of a regular hexagon (Fig i). All wires carry the same magnitude of current in the directions shown. What is the direction of the resultant magnetic field at the centre O of the hexagon?

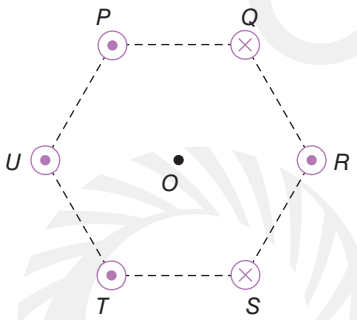


Fig i

A Upwards

B Downwards

C To the left

D To the right

▶ Refer p.198

11 5.2 **HKDSE 2018 Paper 1A Q27**

A straight wire carrying a current I pointing out of the paper is placed in a uniform magnetic field between two pole pieces as shown (Fig j). At which point P, Q, R or S, can the resultant magnetic field be zero? Neglect the effects of the Earth's magnetic field.

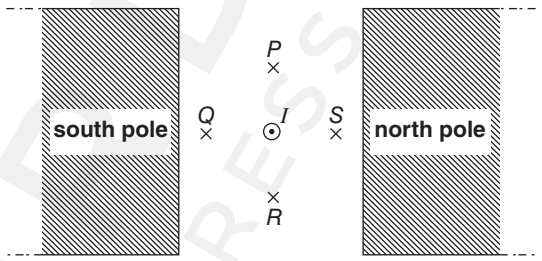


Fig j

A P

B Q

C R

D S

12 5.2 **HKDSE 2021 Paper 1A Q25**

In Figure k, PQR is a straight line with PQ = 10 cm and QR = 5 cm. A long straight wire carrying a current of 0.3A (directed into the paper) is placed at P. When another long straight wire carrying a current I is placed at Q, the resultant magnetic field at R becomes zero. Determine the direction and magnitude of I.

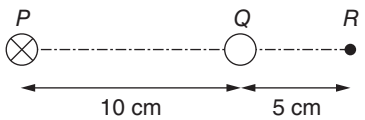


Fig k

	Direction of I	Magnitude of I
A	into the paper	0.1 A
B	into the paper	0.9 A
C	out of the paper	0.1 A
D	out of the paper	0.9 A

Conventional questions

★13 5.2 Two long straight wires carrying an equal size of current are arranged at right angles to each other (Fig l). Points *P*, *Q*, *R* and *S* are situated at the vertices of a square of side 8 cm. All these points are at equal distance from the wires.

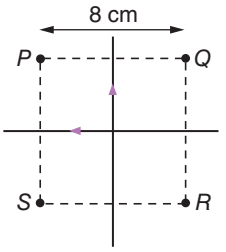


Fig l

(a) State the direction of the resultant magnetic field at each point. (4 marks)

(b) If the current flowing in each wire is 0.5 A, find the strength of the resultant magnetic field at *Q*. (2 marks)

★14 5.2 *X* and *Y* are infinitely long straight wires carrying the same current *I* in opposite directions (Fig m).

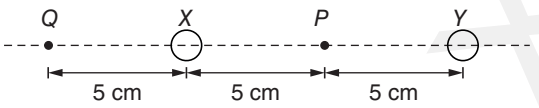


Fig m

(a) The magnetic field at point *P* is 0.02 mT pointing upwards.

(i) Mark the directions of the current in *X* and *Y*.

(ii) Find the current *I*. (3 marks)

(b) Find the strength and direction of the magnetic field at *Q*. (2 marks)

★15 5.2 You are given a metal slinky spring, an ammeter, a rheostat, a magnetic field sensor and a metre rule.

(a) Write the procedures to investigate the relationship between the strength of the magnetic field inside a long solenoid and the number of turns per unit length of the solenoid. Put down your expected results for the experiment. (5 marks)

(b) State any precaution(s) needed. (2 marks)

★16 5.2 Alex designs an electromagnetic door lock (Fig n). A piece of iron is installed in a door. When switch *S* is open, the door is locked due to the attraction of electromagnet *A*. The coils in both electromagnets *A* and *B* are wound on soft-iron cores.

When *S* is closed, points *P* and *Q* will be brought into contact and Alex expects the door can be opened.

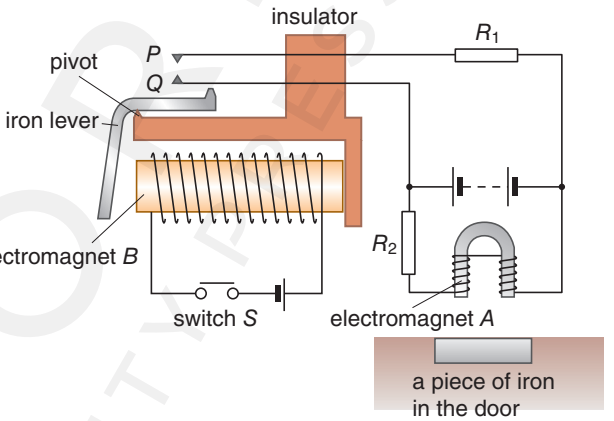


Fig n

(a) (i) Alex found that the door is still locked even when *S* is closed. What is wrong with the circuit? Explain by stating the change in current in each branch of the circuit. (2 marks)

(ii) He decided to fix the problem by moving resistor R_1 to another position in the circuit. Where should he put R_1? Explain how this change works. (3 marks)

(b) The coil in electromagnet *B* is 2 cm long and has 150 turns. Its resistance is 2 Ω. It is connected to a 1.5-V battery. Suppose its core is removed.

(i) Estimate the strength of magnetic field produced at the centre of the coil when the switch is closed. (3 marks)

(ii) What assumption is made in (b)(i)? (1 mark)

(iii) What is the effect of a soft iron core in an electromagnet? (1 mark)

(c) Suggest two methods to make electromagnet *A* stronger. (2 marks)

★17 5.2 In a recycling plant, iron and steel can be taken out using an electromagnet. Kelvin designs the following circuit for this function (Fig o).

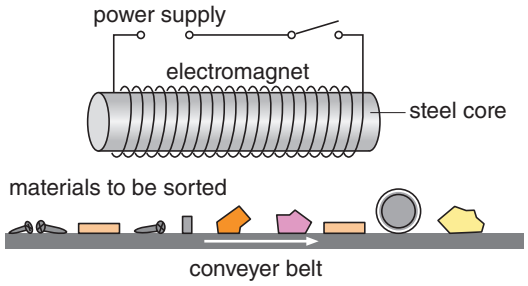


Fig o

He expects that magnetic materials would be attracted by the electromagnet and thereby sorted or picked out.

(a) The electromagnet in this design is inefficient in attracting magnetic materials. Explain why and suggest a way to improve the design. (2 marks)

(b) The electromagnet retains its magnetism even after the switch is opened. Explain why and suggest a solution to this problem. (2 marks)

(c) Suggest two ways to increase the strength of the electromagnet. (2 marks)

★18 5.2 You participate in a competition to make the best electromagnet to transfer paper clips from one container to another without touching them with your hands. The magnet that can transfer the most paper clips will be the winner. You can choose one component from each group of materials in Figure p to make your electromagnet.

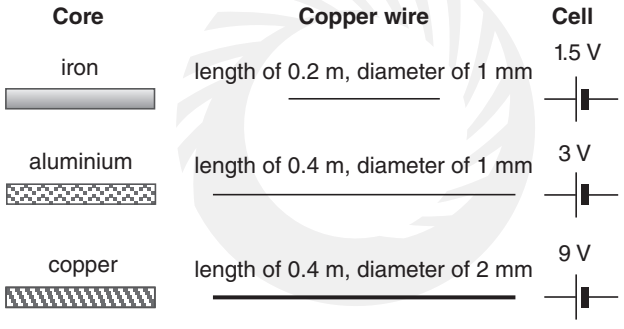


Fig p

(a) What is the best choice from each group? Explain briefly. (6 marks)

(b) Draw a labelled diagram to illustrate your design. (2 marks)

★19 5.2 Figure q shows the schematic diagram of a doorbell. The rod PQ is made of a soft-iron rod and a plastic rod joined together. A spring is attached to the end P of the rod.

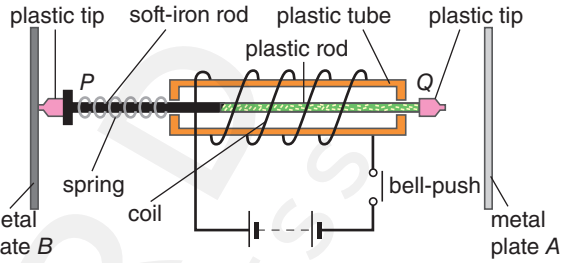


Fig q

(a) Briefly explain how a 'ding-dong' sound is heard when the bell-push is pressed and released. (3 marks)

(b) Explain whether the doorbell will work differently if each of the following changes is made.

(i) The polarity of the battery is reversed.

(ii) The soft-iron rod is replaced with a copper rod.

(iii) The coil is made of thicker wires. (6 marks)

★★20 5.2 Two parallel infinitely long wires A and B are placed 3 cm apart, carrying currents I_A and I_B respectively (Fig r). I_A flows upwards. Point P is at a distance r from A. Both wires and point P are in the same plane.

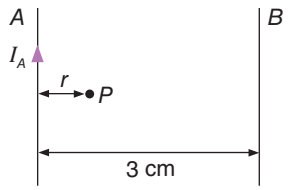


Fig r

(a) The magnetic field at P is zero.

(i) Determine the direction of I_B. (1 mark)

(ii) Find the ratio $\frac{I_A}{I_B}$. Express your answer in terms of r. $\frac{r}{0.03-r}$ (2 marks)

(b) If I_A is increased by 3.5 A, the strength of the magnetic field at P becomes 1.4×10^{-4} T.

(i) Show that $r = 0.005$ m. (2 marks)

(ii) Find the field direction at P. (1 mark)

Refer p.205

21 **HKDSE 2020 Paper 1B Q3**
5.2

Read the following passage about a **magnetically levitated (maglev) train** and answer the questions that follow.

'A maglev train car is just a box with magnets on the four corners,' says Jesse Powell, the son of the maglev train inventor. The electromagnets employed have superconducting coils (i.e. coils with extremely low resistance). They therefore can generate magnetic fields 10 times stronger than ordinary electromagnets, enough to levitate and propel a train.

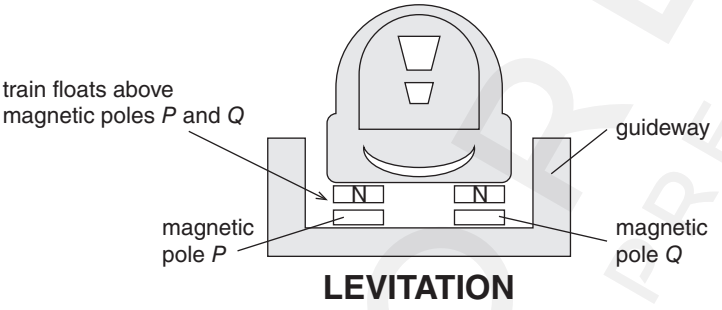


Fig s

Two sets of magnetic fields are set up for different functions. One is to make the train float a few centimetres above magnetic poles *P* and *Q* as shown while the other is a propulsion system run by an alternating current for moving the train car along the guideway by magnetic attraction and repulsion. This floating design enables a smooth movement of the train. Even when the train travels up to 600 km per hour, passengers inside experience less vibration than travelling on traditional trains.

(a) Explain why electromagnets employing superconducting coils can produce much stronger magnetic fields. (2 marks)

(b) State the polarities of the magnetic poles *P* and *Q* and explain how this arrangement enables the train to float. (2 marks)

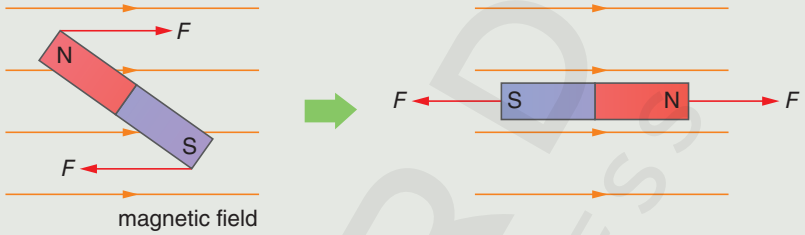
(c) Referring to the resistive forces experienced by the train, explain why a maglev train ride is (i) smoother and (ii) faster. (2 marks)

Exam Report: This question tested candidates' knowledge and understanding in Electromagnetism in the context of magnetically levitated trains. The overall performance was fair. In (a), most candidates were able to point out the lowered resistance allowed a larger current flow. However, a number of them described the current as eddy currents. A considerable number of candidates wrongly believed that 'unlike poles repel each other' in (b) and a few used 'positive pole/charge' to describe magnetic poles. Candidates' performance in (c) was fair. Although most were able to explain the situation in general terms, their descriptions were far from precise. Some simply wrote 'no friction' instead of no friction between the train and the track. Few pointed out explicitly that 'air resistance' was the only resistive force that needed to be overcome.

Know more

Compasses (p.188)

A compass needle is a bar magnet that can rotate. If the magnet does not align with a magnetic field, the magnetic forces acting on its poles will make it turn. It remains at rest only when it aligns along the field.

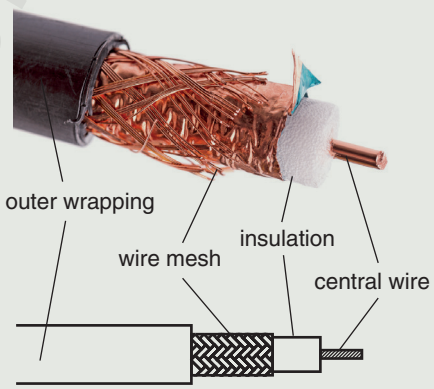


The first compass was invented by the Chinese around 100 AD (late *Han dynasty*). It was called *Sinan*. It consisted of a magnetic spoon, which could rotate on a metal plate. The handle was the S-pole and pointed south as the spoon came to rest.



Coaxial cable (p.205)

A coaxial cable has a central wire for transmitting electrical signals. The wire is first surrounded by an insulating layer, and then all are enclosed in a conducting layer which is usually a copper or aluminium mesh. The combination is again enclosed in a layer of insulation. The wire mesh carries a current in a direction opposite to that in a central wire. In such an arrangement, the resultant magnetic field outside the cable is kept zero and therefore does not affect the circuits nearby.



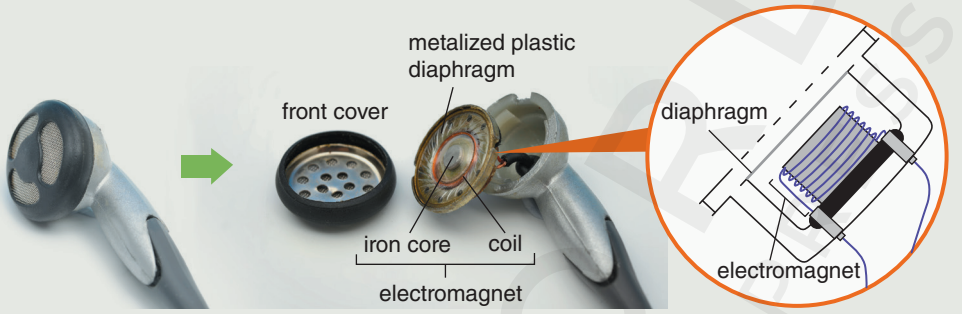
Han dynasty 漢朝 Sinan 司南

Know more

Applications of electromagnets (p.203)

Earphones

An earphone consists of an electromagnet and a thin diaphragm made of metalized plastic. Sound signals are transmitted to the earphone as a varying current, causing the electromagnet to attract the diaphragm with varying magnetic force. As a result, the diaphragm vibrates to produce sound waves.



Maglev trains

The maglev train can travel at a speed above 500 km h^{-1}. As a comparison, the maximum speed of the Airport Express Line in Hong Kong is 135 km h^{-1}.

The maglev train uses a system of electromagnets for levitation and propulsion. The train has no contact with the rail, so it moves with very little friction and hence very high speeds.

To make the train float above the rail, a suitable amount of current is passed through the support electromagnets on the train body and the electromagnets are attracted to the stator on the guideway by an upward magnetic force just balancing the weight of the train.

The guidance electromagnets and the guidance rail are used for stabilizing the train. The repulsion between them can prevent the train from hitting the guideway when moving.

